BASELINE PROJECT PLAN (BPP)

DEVELOPMENT OF AN INTEGRATED FARM MANAGEMENT INFORMATION SYSTEM

Course: INSY281 – System Analysis & Design

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Submission Deadline: 4 November 2026

INTRODUCTION AND PROJECT BACKGROUND

Smart farm is an imaginary medium-sized agricultural enterprise involved in crop production, livestock keeping, and the sale of agricultural products. The farm currently relies on manual records and separate spreadsheets to manage crop production, livestock activities, inventory, sales, and financial transactions. This creates difficulties in retrieving information, monitoring expenses, tracking stock, and preparing accurate management reports.

The proposed Integrated Farm Management Information System (IFMIS) will provide a centralized computerized platform for managing farm operations. It will integrate crop production, livestock management, inventory, sales, expenses, and reporting within one database. Authorized users will be able to enter, update, retrieve, and analyze farm information efficiently.

Problem Statement

The current manual recordkeeping approach leads to duplicated records, inconsistent information, delays in report preparation, difficulties tracking farm inputs, and limited visibility into operational costs and profits. These challenges may contribute to wastage, poor planning, and inefficient resource allocation. The farm therefore requires an integrated system that centralizes records and provides timely, reliable information.

Aim and Objectives

Aim: To analyze, design, develop, test, and implement an Integrated Farm Management Information System that improves farm operations and decision-making.

1. Create a centralized database for crop, livestock, inventory, sales, and financial records.
2. Enable authorized users to record and retrieve farm information efficiently.
3. Generate production, inventory, sales, and financial reports.
4. Introduce role-based access controls and regular data backups.
5. Reduce routine report preparation time by at least 50% against a baseline measured during requirements analysis.
6. Train designated farm employees before implementation is completed.

SYSTEM TYPE, SCOPE AND REQUIREMENTS

System Type

The proposed system is primarily a **Management Information System (MIS)** because it processes operational data into summarized reports that support managerial planning, monitoring, and decision-making. It will also include Transaction Processing System (TPS) functions for recording routine sales, purchases, and expenses.

Scope

The system will include user management, crop production records, livestock records, inventory management, sales and expense recording, report generation, and data backup. Intended users include farm managers, workers, accountants, and authorized administrators. The initial version will exclude automated machinery control, satellite-based crop monitoring, and advanced AI forecasting.

Functional Requirements

1. Register users and assign access permissions.
2. Record crops, planting dates, farm activities, harvest dates, and quantities.
3. Maintain livestock identification, feeding, health, and production records.
4. Record farm inputs, purchases, available stock, and stock movements.
5. Record sales, customer details, operating expenses, and income.
6. Generate daily, monthly, and annual operational and financial reports.
7. Search, edit, and retrieve records according to user permissions.
8. Generate low-stock notifications using predefined inventory thresholds.
9. Maintain audit records of important user activities.
10. Support regular database backups and authorized data recovery.

Non-Functional Requirements

Security: authentication and role-based access; **Performance:** common reports should load within five seconds under normal conditions; **Reliability:** accurate and consistent records; **Usability:** a simple interface supported by training; **Maintainability:** support for updates and corrections; **Data protection:** handling personal information in accordance with applicable requirements.

SYSTEM DEVELOPMENT APPROACH

The project will use the **Systems Development Life Cycle (SDLC) with a Waterfall approach**. This sequential approach is suitable where initial requirements can be documented and approved before development. The stages are:

1. **Planning:** identify the problem, objectives, scope, stakeholders, budget, and schedule.
2. **Requirements analysis:** interview farm managers and employees, observe current processes, and document requirements.
3. **System design:** prepare database schemas, entity-relationship diagrams, interface designs, process models, and security specifications.
4. **Development:** implement the database, interfaces, business logic, and reports.
5. **Testing:** conduct functional, integration, security, usability, and user acceptance testing.
6. **Implementation:** install the system, migrate verified records, train users, and hand over the solution.
7. **Maintenance:** correct defects, monitor performance, provide support, and introduce approved improvements.

A change-control process will assess significant changes to approved requirements, costs, and schedules.

FEASIBILITY ANALYSIS

Feasibility area	Assessment
Economic	Estimated cost is KSh 1,300,000. Expected benefits include reduced paperwork, faster reporting, improved stock control, fewer recordkeeping errors, and better accountability. Feasibility depends on confirming costs and expected savings.
Operational	The system addresses current recordkeeping challenges. User acceptance, management support, employee involvement, and training will be essential.
Technical	The system can use commonly available computers, database software, development tools, and networking. Skills, hosting, backups, and maintenance arrangements must be confirmed.
Schedule	A six-month schedule from January to June 2027 is proposed. Weekly monitoring and milestone reviews will help identify delays.
Legal and contractual	The project will observe applicable Kenyan laws, including the Data Protection Act, 2019 where personal data is processed. Software licenses, hosting agreements, access, and retention arrangements will be reviewed.

Political and organizational	Management support is necessary. Employee resistance and competing stakeholder interests will be addressed through consultation, communication, defined responsibilities, and training.

Overall assessment: The project appears feasible in principle, subject to confirmation of funding, technical resources, stakeholder support, legal compliance, and the proposed schedule.

5.0 DELIVERABLES AND MILESTONES

Deliverables will include: (1) approved project plan and requirements specification; (2) system design documentation, database design, and interface prototypes; (3) a functioning farm management system; (4) a configured and tested database with verified sample or migrated records; (5) a testing report; (6) user documentation, training materials, and training sessions; and (7) implementation, backup, maintenance, evaluation, and handover documentation.

Major milestones are approval of requirements, completion of system design, completion of core development, successful user acceptance testing, completion of training, and formal handover.

PROJECT TIMELINE

Activity	Period	Milestone
Planning and initiation	January 2027	Project plan approved
Requirements analysis	February 2027	Requirements approved
System design	March 2027	Design completed
System development	April 2027	Core features completed
Testing and corrections	May 2027	User acceptance testing completed
Implementation and training	June 2027	System handed over

The project manager will review progress weekly and compare each milestone with the approved schedule. Significant delays will be documented with their causes, effects, and corrective actions.

PROJECT BUDGET

Budget item	Estimated cost (KSh)
Requirements analysis and project planning	120,000
System design and database development	180,000
Software development	350,000
Computers and related equipment	250,000
Hosting, networking, and security	100,000
Testing and quality assurance	100,000
Training and documentation	60,000
Maintenance and contingency allowance	140,000
TOTAL ESTIMATED BUDGET	1,300,000

The estimate exceeds the required minimum of KSh 1,000,000. These are illustrative planning estimates for an imaginary project, not supplier quotations. Actual costs will be confirmed during planning, and expenditure will be monitored against the approved budget.

STAKEHOLDERS, REPORTING AND RISK MANAGEMENT

Stakeholders

Key stakeholders include the farm owner or project sponsor, farm manager, farm employees, accountant, systems analyst, software developers, database administrator, and system users. Their responsibilities will be defined at project initiation.

Progress Reporting

The project manager will prepare weekly status reports for the project sponsor. Reports will cover completed tasks, upcoming activities, milestone progress, expenditure, risks, and corrective actions. Formal reviews will take place at major milestones. Requirements or budget changes will be documented and submitted for approval.

Risk Management

Risk	Mitigation
Resistance to the new system	Involve employees early and provide practical training.
Data loss or corruption	Maintain regular backups and test recovery procedures.
Budget overruns	Monitor expenditure and control scope changes.
Project delays	Review weekly progress and address bottlenecks promptly.
Unauthorized access	Use authentication, access controls, and audit logs.
Inaccurate migrated records	Verify and clean existing data before migration.

EXPECTED OUTCOMES AND CONCLUSION

The system is expected to centralize farm records, improve information accuracy and accessibility, reduce routine reporting time, improve inventory monitoring, and support better financial and operational decisions. Success will be evaluated through functional testing, user acceptance, achievement of milestones, adherence to the approved budget, completion of training, and measurement of the targeted 50% reduction in report preparation time against the established baseline.

In conclusion, the proposed project provides a structured response to Smart farm’s recordkeeping challenges. This Baseline Project Plan defines the objectives, scope, development approach, feasibility, deliverables, budget, schedule, stakeholder responsibilities, reporting procedures, and risk controls required to guide the project.